



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

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SECP3213 - BUSINESS INTELLIGENCE

SECTION 01

RETAIL SALES AND INVENTORY MANAGEMENT ANALYSIS

PHASE 3: Visualization & Dashboard

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Dashboard 1: Sales Performance Overview

Figure 1 is the screenshot of the dashboard for the Sales Performance Overview.

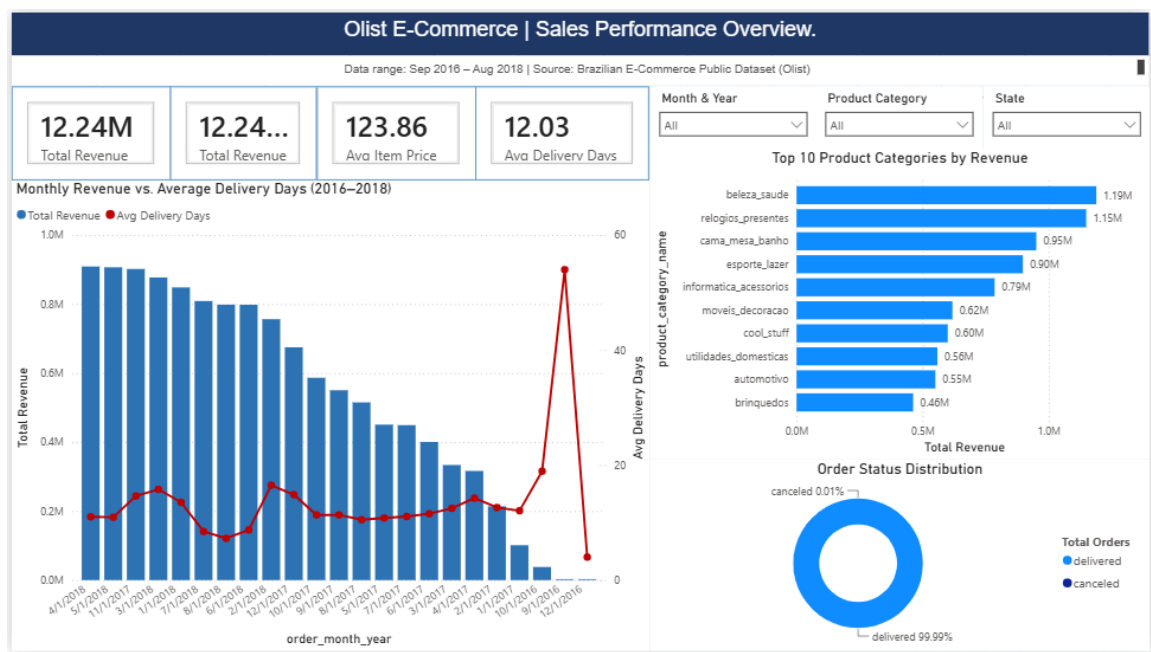


Figure 1: Dashboard for Sales Performance Overview

Dashboard 2: Logistic & Supplier Operations

Figure 2 is the screenshot of the dashboard for the Logistics & Supplier Operations Overview.

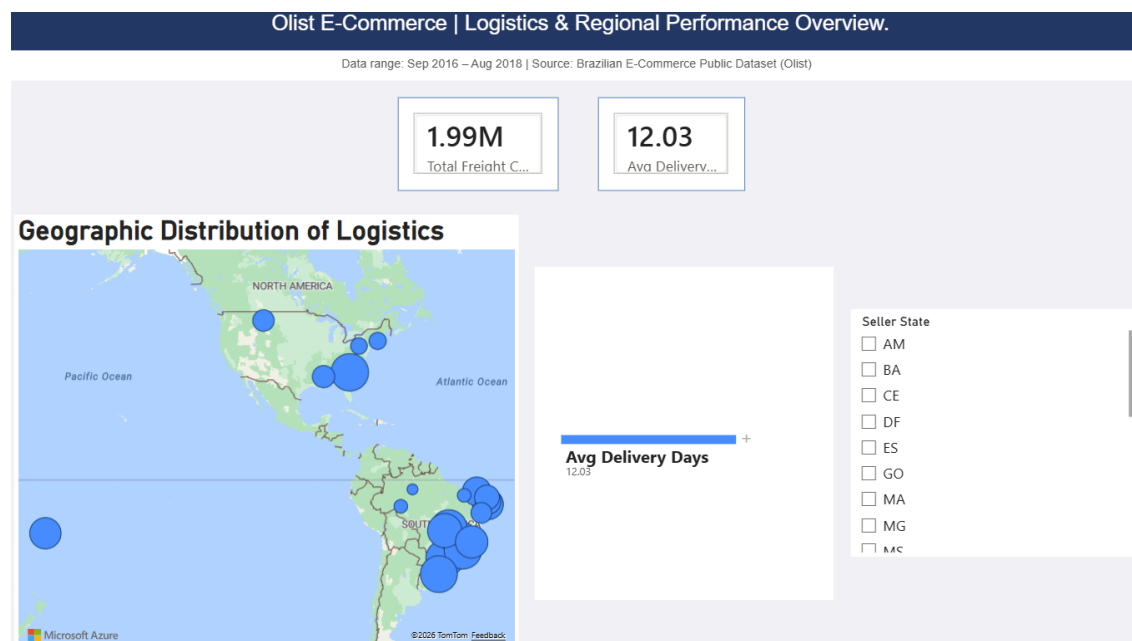


Figure 2: Dashboard for Logistics & Supplier Operations

Storyboard: The Olist

Figure 3 shows the screenshot of the storyboard of the Olist Growth Story 2017.

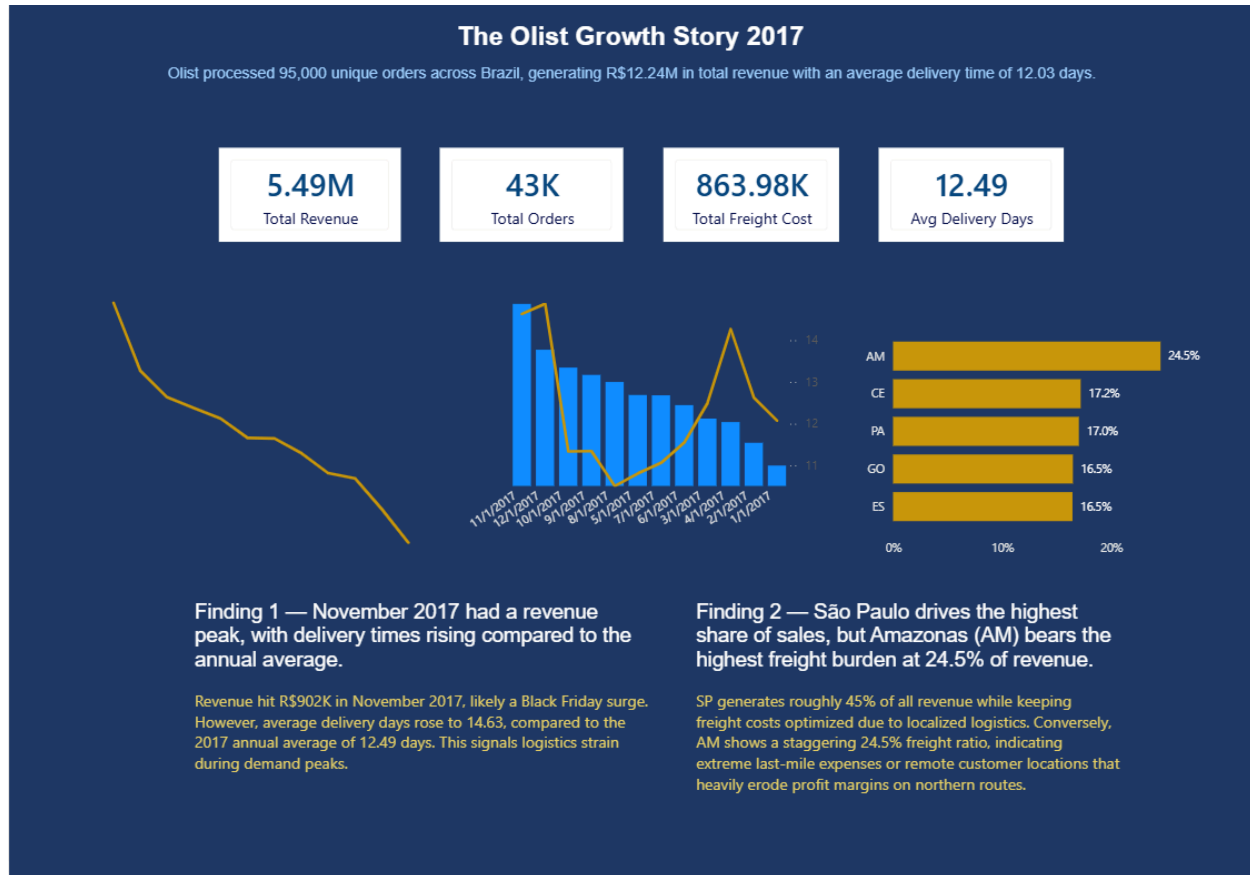


Figure 3: Dashboard for the storyboard of the Olist Growth Story 2017

Key Insights

The insights for both dashboards were derived from analysis of 112,650 order records in the span of September 2016 to August 2018 as in the following:

1. Sales Performance Insights

- The revenue in 2018 reached R\$6.70M in just 8 months, where it surpassed 2017 that reached R\$5.49M. This shows an increase compared to the previous year, which aligns with the swift growth of the marketplace at Olist.

- b. Health & Beauty (beleza_saude) and Watches & Gifts (relogios_presentes) each have a total revenue of R\$1.19M and R\$1.15M respectively make up 19.12% of the total revenue. Especially Health & Beauty products that have the highest average item price at R\$134.59 that shows a high value priority category.
- c. 99.99% of the products with 95,138 orders were successfully delivered, only 0.01% of the orders were cancelled. This indicates a high reliable fulfillment process and a strong accountability on OList platform.

2. Logistic & Supplier Insights

- a. The map visualization shows that São Paulo (SP) contributes the largest share of total freight costs, costing R\$1.99M followed by states such as Minas Gerais (MG) with total freight costs R\$1.97M and Rio de Janeiro (RJ) with total freight costs R\$1.93M. This indicates that logistics operations are heavily centralized in these regions, resulting in higher transportation expenses and greater dependence on a few major seller hubs.
- b. The dashboard reports an average delivery duration of approximately 12 days. While deliveries are being completed successfully, there is potential to improve logistics efficiency and reduce delivery times to enhance customer satisfaction.

3. Olist Growth Story of 2017

- a. Peak Demand Logistics Strain (Finding 1): During November 2017, on Black Friday, Peak Demand Logistics's revenues reached R\$902K. This was a high order volume that put a strain on the logistics network, resulting in a rise in average delivery times to 14.63 days, up from the annual average of 12.49 days.
- b. Regional Freight Burden (Finding 2): São Paulo generates the largest proportion of sales, while having low and optimized freight costs. On the other hand, Amazonas (AM) has the largest freight burden with shipping costs accounting for 24.5% of its revenue, reflecting its significant last mile inefficiency on long-distance transportation.

Business Recommendations

Based on the insights derived from the Sales Performance Overview Dashboard and the Logistics & Regional Performance Dashboard, several data-driven recommendations are proposed to improve business performance and operational efficiency.

1. Sales Performance Dashboard Recommendations

- a. Since Health & Beauty (beleza_saude) and Watches & Gifts (relogios_presentes) both generate 19.12% of all revenue, these products are categorized in an important category and should be given priority. This means that sellers of these categories need to be kept at above a certain stock level.
- b. Aim to limit the reliance only on top 10 product categories, avoid having too many items in the top 10 categories. If there is a decline in demand in any category, there will be a drop in the overall revenue.

2. Logistic & Supplier Insights Dashboard Recommendations

- a. Creating additional fulfillment hubs in strategic regions can reduce shipping distances, lower freight costs, and improve delivery speed for customers located far from major seller states.
- b. Using historical sales and logistics data, the company can better forecast demand and allocate resources during peak periods, reducing the risk of delivery delays.
- c. States with the highest freight costs should be continuously monitored through the dashboard to identify inefficiencies and support data-driven decisions for logistics improvement.

3. Olist Growth Story of 2017

- a. Employ the seasonal capacity planning by carrying out flexible carrier contracts and temporary sorting hubs prior to major sales peaks, such as November, to avoid delivery delays.
- b. Optimize frontier route shipping rates. In certain states, such as AM and CE, freight ratios are high, so it should consider changing the shipping rates to remote areas and negotiating volume discounts with regional carriers when freight ratios are above 17%.